MIL-C-60205A(MU)
1 December 1967
SUPERSEDING
MIL-C-60205(MU)
28 January 1965

MILITARY SPECIFICATION
CARTRIDGE, IMPULSE: M173

1. SCOPE

1.1 This specification covers Cartridge, Impulse: M173 intended for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-60539	- Anodic Coating, Conventional, for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Data Card Information
MS 28741	- Hose Assembly, Detachable End Fitting, Medium Pressure

FSC 1377

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DRAWINGS

Munitions Command

D10522651	- Cartridge, Impulse: M173, Assembly
C8837884	- Carton, Packing, Ammunition for Cartridge, Impulse: M173
C8837885	- Box, Packing, Ammunition for Cartridge, Impulse: M173
IEL-10522651	- Inspection Equipment List for Cartridge, Impulse: M173

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.- Protective finish shall be in accordance with MIL-C-60539 and shall apply to all surfaces of the cartridge regardless of contour.

3.3 Vibration.- The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, propellant or igniter or derangement of any component.

3.4 Primer sensitivity.- The primer shall be fired when drop tested at an energy level of 18 inch ounces in a sectioned cartridge void of igniter and propellant charge. Primer and firing plug area must be free of sealant.

3.5 Functional.

3.5.1 The cartridge shall not fail to fire.

3.5.2 Ballistics.- The cartridge when tested in an M87 Initiator, shall produce the pressures and ignition delay at the temperatures specified in Table I.

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3.5.2 (Cont'd)

TABLE I

Temperature Degrees Fahrenheit (°F)	Pressure 1/ Pounds per Square Inch (PSI) Minimum	Maximum	Ignition Delay Milliseconds Maximum (See 6.1)
-65° to -70°	500	1500	65
65° to 75°	575	1500	65
160° to 165°	625	1500	65

1/ Pressure values shall be produced in a one cubic inch pressure station at the end of 15 feet of MS-28741, hose, Size No. 4.

3.5.3 Carbon monoxide gas.- The cartridge when assembled into an accepted, unused M87 initiator, temperature conditioned at -65° to -70°F, 65° to 75°F and 160° to 165°F and fired into a sealed total volume of 2.58 cubic inches shall not produce carbon monoxide gas leakage greater than 6.0 cubic centimeters total. Volume of gas leakage shall be recorded at 10 minutes and 60 minutes after firing.

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following.

3.6.1 Metal defects.- The component shall be free from cracks, splits, inclusions, porosity or other metal defects.

3.6.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 23 components for protective finish tests, 50 components for hardness tests, 50 cartridges for primer sensitivity tests, 30 cartridges for ballistic tests and 15 cartridges for carbon monoxide gas tests and to be tested in accordance with Table II, and 10 sets of unassembled components (including explosive components) for dimensional and visual examination.

TABLE II

Paragraphs				Type of Test	
Requirement	Test Procedure	Test	Sample Size	Dest.	Non-Dest.
3.2	4.4.1	Protective Finish	23	X	-
Dwg.	4.4.2	Hardness	50	-	X
Dwg.	4.4.3	Airtightness	Each	-	X
3.3	4.4.4	Vibration	12 from Ballistic Sample	-	X
3.4	4.4.5	Primer Sensitivity	50	X	-
3.5.2	4.4.6	Ballistics	30	X	-
3.5.3	4.4.7	Carbon Monoxide Gas	15	X	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.- After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.- The sample for ballistic tests shall be temperature conditioned and conform to requirements as indicated in Tables I and III.

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4.2.1.2 (Cont'd)

TABLE III

Temperature (°F)	Vibrated Cartridges	Not Vibrated
-65° to -70°F	4	6
65° to 75°F	4	6
160° to 165°F	4	6

4.2.1.3 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions

4.3.1 Lot

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- Unless otherwise specified by the procuring agency, a lot shall consist of 1200 cartridges, maximum, and shall consist of:

a. Components produced by a homogeneous process

b. Propellant from not more than one lot

c. Primers from not more than one lot

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic), the criteria in Table IV shall be used. For total defects within a major or minor class the criteria in Table V shall be used. All non-conforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber "O"-rings shall be in accordance with MIL-STD-413. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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4.3.2 (Cont'd)

TABLE IV

Lot Size	Sample Size	Major Accept	Major Reject	Minor Accept	Minor Reject
1 - 300	100% Examination				
301 - 1,300	110	1	2	2	3
1,301 - 3,200	150	2	3	3	4
3,201 - 8,000	225	3	4	4	5
8,001 - 22,000	300	4	5	5	6

TABLE V

Total Allowable Defects per Sample Size

Number of Defects in Class	Sample Size for Major Defects	110	150	225	300
2 - 3		1	2	3	4
4 - 7		2	3	4	5
8 - 19		3	4	5	7
20 - 43		4	5	8	10
Over 43		6	8	11	14

Number of Defects in Class	Sample Size for Minor Defects	110	150	225	300
2 - 3		2	3	4	5
4 - 7		3	4	5	7
8 - 16		4	5	8	10
17 - 48		6	8	11	14
Over 48		8	11	17	20

4.3.2.1 Case, Cartridge, Drawing 8597231, covering a detail of Drawing D10522651

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual

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4.3.2.1 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Concentricity of head with body	Gage
102. Straightness of base	Gage
103. Diameter of head, min.	Gage
104. Diameter of body, min.	Gage
105. Thickness of wall	Gage
106. Thickness of wall at base	Gage
107. Protective finish inadequate or defective	Visual
108. Burr or sharp corner	Visual
109. Foreign matter	Visual
MINOR	
201. Diameter of head, max.	Gage
202. Diameter of body, max.	Gage
203. Total length, max.	Gage
204. Location of datum diameter	Gage
205. Radius missing or incorrect	Visual
206. Surface finish improper	Visual

4.3.2.2 Plug, Firing, Drawing 8597234, covering a detail of Drawing D10522651

CRITICAL

None defined

Defect	Method of Inspection
MAJOR	
101. Concentricity of flange with body	Gage
102. Diameter of flange	Gage
103. Diameter of body, min.	Gage
104. Height of boss, min.	Gage
105. Metal defective	Visual
106. Surface finish of body, improper	Visual
107. Burr or sharp edge	Visual
108. Foreign matter	Visual

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter of body, max.	Gage
202. Total length	Gage
203. Length of body	Gage
204. Radius missing or incorrect	Visual
205. Surface finish improper, except as otherwise classified	Visual

4.3.2.3 Ring, Retaining, Drawing 8597235, covering a detail of Drawing D10522651

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity inside diameter with outside diameter	Gage
102. Diameter inside, max.	Gage
103. Diameter outside, min.	Gage
104. Protective finish inadequate or defective	Visual
MINOR	
201. Diameter inside, min.	Gage
202. Diameter outside, max.	Gage
203. Total length	Gage
204. Radius	Visual
205. Surface finish improper	Visual

4.3.2.4 Primer M29A1, Drawing C8798312, covering a detail of Drawing D10522651

CRITICAL	
1. Improper assembly	Visual
2. Any part missing	Visual

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
CRITICAL (Cont'd)	
3. Primer pellet broken, cracked or dislodged	Visual
4. Metal defective	Visual
5. Primer foil discolored	Visual
6. Mixed primers	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.5 Head, Cartridge, Drawing C10522652, covering a detail of Drawing D10522651

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of forward shoulder with primer pocket	Gage
102. Concentricity of flange with primer pocket	Gage
103. Concentricity of body with primer pocket	Gage
104. Concentricity of large counterbore with primer pocket	Gage
105. Concentricity of flash hole with primer pocket	Gage
106. Diameter of primer pocket	Gage
107. Diameter of small counterbore, max.	Gage
108. Diameter of large counterbore, max.	Gage
109. Depth of primer pocket	Gage
110. Location of flange	Gage
111. Protective finish inadequate or defective	Visual
112. Surface finish of diameter of primer pocket and small counterbore	Visual
113. Burr, except as permitted	Visual
114. Foreign matter	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter of small counterbore, min.	Gage
202. Diameter of large counterbore, min.	Gage
203. Diameter of body at base	Gage
204. Diameter of flange	Gage
205. Diameter across step at base of angle	Gage
206. Diameter across flat	Gage
207. Depth to base of small counterbore	Gage
208. Depth of small counterbore	Gage
209. Total length	Gage
210. Location of datum diameter	Gage
211. Location of shoulder	Gage
212. Radius or chamfer missing or incorrect	Visual
213. Surface finish improper, except as otherwise classified	Visual

4.3.2.6 Cup, Retainer, Drawing B10522653, covering a detail of Drawing D10522651

CRITICAL	
1. Diameter, small	Gage
2. Metal defective	Visual
MAJOR	
101. Concentricity of counterbore with bore	Gage
102. Concentricity of small diameter with bore	Gage
103. Concentricity of large diameter with bore	Gage
104. Diameter, large, min.	Gage
105. Diameter of bore	Gage
106. Surface finish of bore, counterbore, small and large diameter, improper	Visual
107. Burr or sharp corner	Visual
108. Foreign matter	Visual

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter, large, max.	Gage
202. Diameter, small	Gage
203. Depth of counterbore	Gage
204. Total length	Gage
205. Length of shoulder	Gage
206. Surface finish improper, except as otherwise classified	Visual

4.3.2.7 Assembly, Cartridge, Impulse: M173, Drawing D10522651

CRITICAL	
1. Firing plug above flush with retaining cup	Gage
2. Primer not properly seated in primer pocket	Gage
3. Metal defective	Visual
4. Any part missing	X-ray 1/ and Visual
5. Improper assembly	X-ray 1/ and Visual
6. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Diameter of body, max.	Gage
103. Crimping of primer missing or incorrect	Visual
104. Foreign matter	Visual
MINOR	
201. Diameter of crimp, min.	Gage
202. Diameter of head, max.	Gage
203. Total length, max.	Gage
204. Location of datum diameter	Gage
205. Varnish over marking missing or inadequate	Visual

1/ The cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

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4.3.2.8 Carton, Packing, Ammunition for Cartridge, Impulse: M173, Drawing C8837884

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual and Visual
202.	Quantity of contents incorrect	Visual
203.	Separator or filler missing	Visual
204.	Heat sealer incomplete, loose or distorted	Visual
205.	Cut or damage through impregnated box	Visual
206.	Marking missing, incorrect or illegible	Visual

4.3.2.9 Box, Packing, Ammunition for Cartridge, Impulse: M173, Drawing C8837885

CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual and Visual
202.	Hardware missing, loose, broken or improperly engaged	Visual
203.	Strapping missing, loose, broken or improperly assembled	Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of cartons incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

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4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.- Twenty three units shall be selected from each anodized lot for the protective finish test. Failure of any unit to comply with the requirements of 3.2 shall be cause for rejection of the lot.

4.3.3.2 Hardness.- Fifty units shall be selected for this test. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.3 Airtightness.- Each unit shall be subjected to the airtightness test. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the unit.

4.3.3.4 Vibration.- Twelve of the 30 assembled, x-rayed cartridges of the initial production sample selected for ballistic tests and properly identified shall be subjected to vibration as in 3.3. After the vibration test, the 12 vibrated cartridges shall be x-rayed as directed in Note 1/ of 4.3.2.7. Evidence of any disintegration of the primer, propellant or igniter or derangement of any components shall be cause for rejection of the lot. When a change occurs in the lot of primers, propellant or igniter after the initiation of production, 12 assembled cartridges from the first production lot shall be tested in accordance with this paragraph.

4.3.3.5 Primer sensitivity.- A sample of 50 assembled cartridges from the initial production sample and 25 cartridges from each subsequent lot shall be tested for sensitivity. Failure of any primer to comply with the requirements of 3.4 shall be cause for rejection of the lot.

4.3.3.5.1 Sealant application.- The sample of cartridges consumed in the primer sensitivity test shall be disassembled and examined for improper sealant application. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.6 Functional.- The following tests shall be performed by the Government:

4.3.3.6.1 Ballistics.- A sample of 30 assembled cartridges from each production lot shall be ballistically tested after temperature conditioning in accordance with Table VI and 4.4.6.1

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4.3.3.6.1 (Cont'd)

TABLE VI

Temperature Degrees (°) Fahrenheit (F)	Sample
-65° to -70°	10
65° to 75°	10
160° to 165°	10

4.3.3.6.2 Carbon monoxide gas.- Fifteen cartridges from the initial production sample and 15 cartridges from each production lot shall be tested for carbon monoxide gas. The cartridge shall be temperature conditioned as specified in Table VII and 4.4.6.1. Failure of any cartridge to comply with the requirements of 3.5.3 shall be cause for rejection of the lot.

TABLE VII

Temperature (°) (F)	Sample Size	
	Initial Production Lots	Production Lots
-65° to -70°	5	5
65° to 75°	5	5
160° to 165°	5	5

4.3.4 Inspection equipment.- The examination and tests shall be performed using equipment listed on IEL-10522651.

4.4 Test Methods and procedures.

4.4.1 Protective finish.- Test procedures for protective finish shall be in accordance with MIL-C-60539.

4.4.2 Hardness.- The hardness test shall be performed on all heat-treated components prior to the application of protective finish. A hardness testing machine shall be used to determine compliance with the drawing requirements.

4.4.3 Airtightness.- The method for determining airtightness shall be by use of the equipment listed on IEL-10522651.

4.4.4 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

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4.4.5 Primer sensitivity.- Remove and discard all powder charges from the test cartridges by sectioning the cartridge case around its periphery 1/2 inch maximum above the base. Drop test 50 sectioned cartridges from the initial production sample and 25 sectioned cartridges from each production lot at 9 inches, using a steel bass 15/16 inches in diameter, weighing 1.94 ounces plus or minus 0.02 ounces in a primer sensitivity machine with a cartridge holder.

4.4.5.1 Sealant application.- Upon completion of the primer sensitivity test, each cartridge shall be disassembled, exposing the firing plug and primer areas for examination.

4.4.6 Ballistics.- The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned in accordance with 4.4.6.1.

4.4.6.1 Temperature conditioning.- Cartridges shall be loaded into acceptable M87 initiators with end caps to cover the gas ports during the conditioning periods. The temperature of the propellant within the cartridge shall be considered as the conditioning temperature and will be determined by means of a thermocoupler attached to the center of the propellant charge of a specimen cartridge assembled in a standard test device similar to the initiators undergoing tests. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature.

4.4.7 Carbon monoxide gas.- The test sample shall be rigidly supported on the upright member of the fixture. A protective wire mesh cover shall be placed over the bell jar. The sample shall be tested within 10 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample unit shall be reconditioned in accordance with 4.4.6.1.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8837884 and C8837885, except that all markings shall refer to Cartridge, Impulse: M173.

5.2 Data card.- Data card information shall be in accordance with MIL-STD-1167.

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6. NOTES

6.1 Ignition delay is defined as the time that elapses between firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

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Preparing activity:

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